

Samarth Shukla

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Education

Jaypee University of Engineering and Technology, Guna
B.Tech in Computer Science

Sept '22 – May '26

Experience

Software Engineer, Advanced Impactor (Remote)

March '26 – Present

- **Advanced Impactor**

- Engineered **3+ production applications**, including **Advanced Impactor**, **AlFactory**, and **CoachPlex**, using React.js and modern full-stack technologies to deliver scalable, production-ready systems.
- Architected and developed the **Advanced Impactor CRM** using **React + Vite**, integrating **AWS Lambda**, **API Gateway**, and **Amazon S3** for scalable backend APIs and media workflows.
- Built an internal **analytics dashboard** tracking **10+ platform metrics**, including user sign-ups, video uploads, and platform activity, enabling centralized monitoring of application usage.
- Improved web application performance using techniques including **lazy loading**, **code splitting**, **optimized asset loading**, and **API optimization**, reducing unnecessary frontend and network processing.
- Standardized **MongoDB connection configuration across Lambda functions**, applying connection pooling and timeout/idle settings including **30s max idle time**, **pool size of 1**, and **5s server-selection timeout** to address recurring database connection issues in the serverless environment.
- Developed and optimized **4+ AI-powered cricket analytics modules** including Pitch Maps, Wagon Wheels, Shot Maps, and biomechanics using **Python**, **OpenCV**, **YOLO**, and **YOLO Pose**, validating computer-vision workflows across **100+ cricket videos**.

SDE Intern, Aura-AI (Remote)

August '25 – Feb '26

- **Advanced Impactor**

- Engineered **4+ cricket analytics modules**, including Pitch Maps, Wagon Wheels, and Shot Maps, using Flutter and Python.
- Implemented and tested **15+ Flutter features** for video upload, playback, AI analytics, and authentication across Android and iOS.
- Managed the Ubuntu-based AI processing pipeline integrating **AWS S3**, **MongoDB**, and **Python services** for video storage and processing.
- Validated a custom YOLO model with OpenCV across **100+ cricket videos** for automated detection and segmentation.

- **PadhAI**

- Architected a full-stack learning platform with **5+ modules** for class management, assignments, quizzes, and AI-assisted learning.
- Implemented a RAG-based AI assistant supporting contextual PDF Q&A, image-based queries, and persistent chat history.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs

AI/ML: YOLO, OpenCV, LangChain

Frontend: React.js, Next.js, Vite

Databases: PostgreSQL, MySQL, MongoDB, Supabase, Firebase

Tools: Git, GitHub, Docker, Ubuntu, AWS

Technical Projects

Video Voice Modifier

GitHub

- OpenAI Whisper, Google Gemini, Signal Processing, Next.js, FastAPI

- Engineered a full-stack voice modification system supporting video voice conversion across **20+ languages** and **50+ videos**, reducing transcription and dubbing time by **80%** and achieving **>90% lip-sync accuracy**.

Travisco

GitHub

- React.js, FastAPI, Google Gemini, LangChain, Firebase

- Engineered an AI-powered travel platform with monument recognition, contextual Q&A, and intelligent content discovery, reducing page-load time by **50%**, media-upload processing by **60%**, and achieving **92%** filtering accuracy.